

Postpartum Hemorrhage 04: The 4 T's and Cause-Specific Management

Companion reading handout for simultaneous listening.

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The four T's and cause-specific management.

In episode three, we focused on medications.

Oxytocin first.

Tranexamic acid early.

Then choose the next uterotonic by the patient, not by the drawer.

Now we widen the lens

Because the most important question in postpartum hemorrhage is not, "Which medication comes next?"

The most important question is, "Why is she bleeding?"

The four T's are tone, trauma, tissue, and thrombin.

But do not treat them as a memory trick.

Treat them as four doors.

Behind each door is a different treatment pathway.

The mistake is to keep pushing on the tone door when the patient is bleeding behind another one.

The phrase for this episode is:

The four T's are four doors.

Do not keep pushing on the tone door when the patient is bleeding behind another one.

Let us begin where the resident begins: with a hand on the uterus

You have called help.

You have asked for blood loss quantification.

You have started access and labs.

You have begun oxytocin and added tranexamic acid.

Now put your hand on the uterus.

It is either soft or it is not.

That one physical finding changes the next minute.

If the uterus is soft, you are in the tone pathway until proven otherwise.

If the uterus is firm and bleeding continues, you must shift quickly toward trauma, tissue, and thrombin.

The uterus speaks first.

Listen to it.

Door one is tone

Tone means uterine atony, and atony is the most common cause of postpartum hemorrhage.

At term, the placental site receives enormous blood flow. After placental delivery, hemostasis depends on myometrial fibers contracting around the spiral arteries. The uterus becomes the living tourniquet.

Atony means the tourniquet fails.

The bedside picture is familiar.

The uterus is soft.

It may be enlarged or boggy.

Bleeding may come in waves or gushes.

Massage may temporarily improve the tone, and then the uterus relaxes again.

Your treatment is mechanical, pharmacologic, and diagnostic at the same time.

Massage the uterus.

Use bimanual compression when needed.

Empty the bladder.

Insert a Foley catheter.

Start oxytocin.

Add tranexamic acid.

Choose additional uterotonics according to contraindications.

And keep asking whether this is purely tone.

Because tissue can cause tone.

Retained placental fragments can prevent full contraction.

Inversion can present as hemorrhage and a fundus that is not where it should be.

Accreta can turn attempted placental separation into catastrophic bleeding.

So even in the tone pathway, do not become blind to the other doors.

There is also a technique point.

Bimanual uterine massage is not random pressure.

The uterus is compressed between an external hand on the fundus and an internal hand in the vagina.

The goal is effective compression, not aggressive injury.

Gabbe specifically warns that overly aggressive massage can injure large vessels in the broad ligament.

The tone pathway should feel active but controlled.

Soft uterus.

Massage.

Bladder empty.

Oxytocin.

Tranexamic acid.

Second-line uterotonic by contraindication.

If bleeding persists despite medical therapy, escalate.

Before tamponade or advanced intervention, inspect the genital tract for lacerations.

If atony remains the cause, move toward Bakri balloon, embolization when stable and feasible, or surgical intervention.

That brings us to door two: trauma.

Trauma means bleeding from injury.

It includes cervical lacerations, vaginal lacerations, perineal tears, vulvar lacerations, hematomas, uterine incision extensions at cesarean, uterine rupture, and injury to adjacent vessels or organs.

The classic clue is bleeding despite adequate uterine tone.

The uterus is firm.

The blood keeps coming.

That is the moment to stop saying, "Give another uterotonic," and start saying, "Expose and inspect."

Gabbe teaches that genitourinary tract laceration should be suspected when bleeding persists after delivery despite adequate uterine tone.

The evaluation should begin superiorly and move downward.

Cervix.

Vaginal fornices.

Vagina.

Perineum.

Vulva.

You need exposure, lighting, retraction, and pain control.

If you cannot see, you cannot repair.

If the patient is in pain and the field is poor, move to the operating room with anesthesia assistance.

This is not delay. It is how you stop bleeding safely.

Cervical and vaginal fornix lacerations can be difficult because they hide high and retract.

The apex matters.

For a cervical laceration, secure the apex of the tear because it may be the major bleeding source.

A useful technique is to start suturing at the proximal end, then use the suture for traction to bring the distal portion into view.

Perineal trauma is more familiar, but familiarity should not make it casual.

After third- or fourth-degree repair, perform a digital rectal examination to confirm rectal integrity and avoid missed sutures through the rectum.

Now think about hematomas.

Not all bleeding is visible.

A patient may have severe pain, pressure, tachycardia, falling blood pressure, or instability with less external bleeding than expected.

That is a concealed bleeding pattern.

Vulvar hematomas are often visible because fascial boundaries push the swelling toward the skin.

Large vulvar hematomas usually require surgical drainage, evacuation, closure of dead space, pressure dressing, and catheterization until edema improves.

Vaginal hematomas may be less obvious.

Small, nonexpanding vaginal hematomas can sometimes be observed.

Larger or expanding hematomas require surgical intervention, often with packing or tamponade of raw edges. If bleeding persists, embolization may be considered.

Retroperitoneal hematoma is the most dangerous concealed form.

It may remain subtle until the patient is unstable from massive hemorrhage.

It can follow internal iliac branch injury, cesarean hemostasis failure, operative delivery trauma, or uterine rupture.

Treatment often requires laparotomy, evacuation, and arterial ligation, with embolization as a primary or adjunct option in selected stable cases.

So the trauma door sounds like this:

Firm uterus, ongoing bleeding. Expose. Inspect cervix down to vulva. Repair what is bleeding. Do not accept a poor view. Think concealed hematoma when pain or shock is out of proportion.

Door three is tissue

Tissue means retained placenta, retained membranes, retained products of conception, and placenta accreta spectrum.

Retained tissue causes hemorrhage in two ways.

First, it may bleed directly.

Second, it can prevent the uterus from contracting well, so it can look like atony.

That is why tissue and tone often overlap.

Gabbe defines retained products when spontaneous expulsion has not occurred within 30 to 60 minutes after delivery.

Risk factors include midtrimester delivery, chorioamnionitis, and accessory placental lobes.

At the bedside, tissue should come to mind when the placenta is incomplete, membranes are missing, the uterus remains boggy despite treatment, bleeding persists, or the history suggests adherent placenta.

The uterine cavity must be explored when retained tissue is suspected.

Manual exploration is both diagnostic and therapeutic.

Wrapping the examining hand with moist gauze can help remove placental fragments and membranes.

Ultrasound can help if manual access is difficult, maternal body habitus limits examination, pain relief is inadequate, or retained fragments are uncertain.

If retained tissue is confirmed, it must be removed.

Options include manual extraction or curettage.

If bleeding is excessive, curettage belongs in the operating room.

Ultrasound guidance can help confirm complete evacuation.

Nitroglycerin can help manual extraction by rapidly relaxing the uterus.

Gabbe describes nitroglycerin 50 to 200 micrograms intravenously for retained placenta extraction.

This is a controlled use of uterine relaxation for a specific purpose: remove the trapped tissue so the uterus can contract.

But there is a major danger inside the tissue door.

Placenta accreta is not solved by pulling harder.

Say that clearly.

Accreta is not solved by pulling harder.

If the placenta does not separate and accreta is suspected, aggressive manual removal can convert a difficult situation into uncontrolled hemorrhage.

The risk context matters.

Prior cesarean with placenta previa is the classic high-risk combination.

Placenta accreta spectrum includes accreta, increta, and percreta, depending on depth of invasion.

Prenatal diagnosis improves outcomes because accreta is best managed with planned multidisciplinary care.

But sometimes the diagnosis is first suspected at delivery.

If the placenta will not separate easily, especially in a high-risk patient, pause.

Call senior help.

Prepare blood.

Prepare anesthesia and the operating room.

Do not keep pulling in hope.

At cesarean, if accreta is suspected, the safer strategy is often to avoid disturbing the placenta, incise away from it when possible, deliver the fetus, close the hysterotomy, and proceed according to the planned surgical strategy.

Definitive management may include hysterectomy, or in selected cases a conservative approach such as leaving the placenta in situ, embolization, hemostatic sutures, devascularization, or tamponade.

But conservative management carries real risk: delayed hemorrhage, infection, fistula, uterine necrosis, later surgery, hysterectomy, and even death.

For the resident, the memory line is:

Incomplete placenta: think tissue. Trapped placenta: remove carefully. Suspected accreta: stop pulling and call help.

Door four is thrombin

Thrombin means coagulopathy.

It includes inherited disorders, anticoagulants, and acquired consumption of clotting factors.

In the PPH room, the most urgent thrombin problem is usually acquired coagulopathy from massive bleeding or obstetric disease.

Think of placental abruption.

Severe preeclampsia.

HELLP syndrome.

Amniotic fluid embolism.

Sepsis or chorioamnionitis.

Fetal demise.

Massive antepartum or postpartum hemorrhage.

Congenital disorders such as von Willebrand disease.

Anticoagulant exposure.

The clinical clue is oozing.

Bleeding from puncture sites.

Bleeding from raw surfaces.

Diffuse surgical oozing.

Hypotension out of proportion to visible blood loss.

Renal injury.

Acute lung injury.

A patient who looks sicker than the visible bleeding explains.

Laboratory clues include low fibrinogen, low platelets, prolonged prothrombin time, prolonged activated partial thromboplastin time, and evidence of hemolysis when relevant.

Fibrinogen is particularly important in obstetric hemorrhage.

Send it early.

Trend it when bleeding continues.

If timely labs are unavailable, Gabbe describes a rough bedside clot test.

Draw 5 milliliters of maternal blood into an empty red-top tube and observe.

If no clot is visible within 6 minutes, or if a clot forms and then lyses within 30 minutes, fibrinogen is usually below 150 milligrams per deciliter.

This is not a replacement for laboratory-guided care.

It is a rough warning sign when the laboratory answer is not yet available.

The thrombin door has a different treatment logic.

More oxytocin does not correct DIC.

More massage does not replace fibrinogen.

Thrombin problems require treatment of the underlying cause and replacement of consumed components.

That means blood products and clotting factors while source control continues.

Gabbe gives physiologic targets: hematocrit greater than 21 percent, platelets above 50,000 per cubic millimeter, fibrinogen above 100 milligrams per deciliter, prothrombin time and activated partial thromboplastin time less than 1.5 times control, adequate oxygenation, and normothermia.

The local protocol adds the practical Stage three physiology: activate massive hemorrhage, maintain saturation above 95 percent, prevent hypothermia, correct acidosis, and check calcium and potassium frequently during major transfusion.

The thrombin script is:

This is oozing and coagulopathy. Keep source control moving, but replace blood products and clotting factors now. Do not wait for perfect labs if the clinical picture is clear.

Now let us make the four doors into a single bedside flow

You are called to hemorrhage.

The first actions are already moving: help, quantification, access, labs, blood bank, uterine massage, oxytocin, tranexamic acid.

Now your hand is on the uterus.

If it is soft, say:

Tone.

Treat atony.

Massage, empty bladder, uterotonics, tranexamic acid.

If persistent, inspect for trauma and retained tissue, then escalate to tamponade, embolization, or surgery.

If it is firm and bleeding continues, say:

Not just tone.

Inspect for trauma.

Check placenta and uterine cavity for tissue.

Check coagulation for thrombin.

If the placenta is incomplete, say:

Tissue.

Explore the uterus, remove retained tissue, use ultrasound when useful, and do curettage in the operating room if bleeding is excessive.

If the placenta is adherent and accreta is plausible, say:

Stop pulling.

Call help and prepare for surgical hemorrhage.

If the bleeding is diffuse oozing, say:

Thrombin.

Send and repeat coagulation studies, fibrinogen, and platelets.

Activate massive hemorrhage when indicated.

Replace components while treating the cause.

Now let us close with four cases

Case one

After delivery, the uterus is boggy and enlarged. Bleeding improves with bimanual massage but returns when the uterus relaxes.

This is tone.

Treat atony quickly, but do not forget to check whether retained tissue is preventing contraction.

Case two

The uterus is firm. Blood is bright and persistent. There was an operative vaginal delivery.

This is trauma until proven otherwise.

Expose, inspect, and repair. If the field is poor, move to the operating room with anesthesia.

Case three

The placenta is missing a lobe. The uterus is not contracting well. Bleeding continues.

This is tissue, and tissue is feeding tone.

Explore the cavity. Remove retained tissue. Use ultrasound guidance when helpful.

Case four

There is bleeding from puncture sites and diffuse oozing. The patient had severe abruption. Fibrinogen is falling.

This is thrombin.

Replace blood and clotting factors while correcting the cause and maintaining source control.

The four T's are four doors.

Tone is the soft uterus.

Trauma is bleeding despite tone.

Tissue is the incomplete or trapped placenta.

Thrombin is the patient who cannot clot.

Do not keep pushing on the tone door when the patient is bleeding behind another one.

In the next episode, we will move into advanced hemorrhage control: Bakri balloon, embolization, laparotomy, compression sutures, arterial ligation, packing, and hysterectomy. The goal will be to understand when each escalation is appropriate and how to move before the patient has lost the chance to be stabilized.